UNIT 1 · WHY DCF FAILS

The Ornstein-Uhlenbeck Process for Liquidity

Session 4 · Intuition only — full math in Session 25 (Track 2)

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Primary Text: Liquidity Illusion (Forthcoming, 2026)

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What we'll cover today

1. **Why we need a process model**
   Description ≠ prediction
2. **Three properties our model must have**
   Mean reversion · stochastic · bounded variance
3. **The OU process**
   SDE form and verbal translation
4. **What each parameter does**
   $\kappa \cdot \bar{L} \cdot \sigma$ — calibrated values
5. **Self-diagnostic for Track 2**
   If today felt easy, Track 2 may suit you

From data to model: what we have, what we need

WHAT WE HAVE (from Session 3)

- Time series of secondary market discounts 2003–2024
- Implied liquidity state L(t) at quarterly frequency
- Four regimes with distinct statistical character
- Mean / std / autocorrelation by regime

Description is rich. But prediction requires more.

WHAT WE NEED (to value)

- A MODEL of how L(t) evolves through time
- Closed-form properties: mean, variance, autocorrelation
- Conditional distribution: $L(t+h) \mid L(t) = \ell$
- Stationary distribution: long-run L behavior
- Forecast-ability for asset cash flow valuation

Without a process model, we describe the past but cannot value the future.

Three properties our model must have

Property 1

Mean reversion

Discounts widen in crises but return toward a long-run average. The unconditional mean exists and is finite. Rules out random walk.

→ Vasicek, CIR, OU all satisfy this

Property 2

Stochastic shocks

Period-to-period changes are partly random — not deterministic. Cannot be predicted from L(t) alone with certainty.

→ Brownian noise dW captures this

Property 3

Bounded variance

Long-run variance is finite. L(t) does not drift indefinitely. The unconditional distribution is well-defined.

→ Mean reversion + Brownian noise → bounded variance

These three properties together rule out simple Brownian motion (unbounded variance) and rule in the OU family.

The Ornstein-Uhlenbeck process

The SDE (Stochastic Differential Equation):

$$dL_t = \kappa(\bar{L} - L_t)\,dt + \sigma\,dW_t$$

Verbal translation:

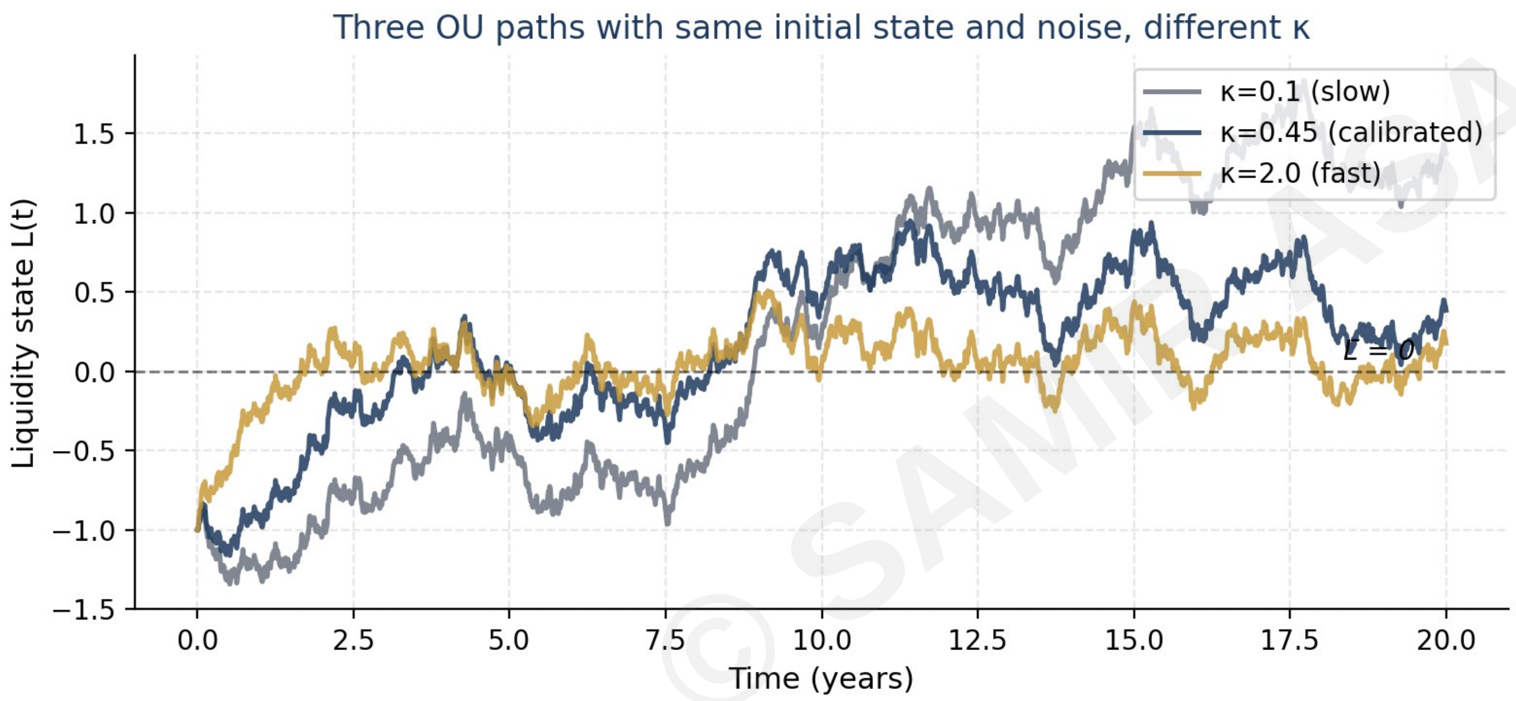
L(t) tends to revert toward $\bar{L}$ at speed κ, with random shocks of magnitude σ.

History

Originally derived by Leonard Ornstein and George Uhlenbeck in 1930 to model the velocity of a particle undergoing Brownian motion in a viscous fluid.

The same mathematics describes any mean-reverting stochastic variable: interest rates (Vasicek), volatility, and our liquidity state.

κ controls mean reversion speed



Half-life formula

$$\text{Half-life} = \frac{\ln 2}{\kappa}$$

Time to revert halfway to $\bar{L}$.

κ	Half-life
κ = 0.1	6.9 years
κ = 0.45	**1.5 years**
κ = 2.0	0.35 years

Calibrated κ = 0.45 (1.5-yr half-life) matches the secondary market data.

Analogy: radioactive decay uses the same half-life math.

$\bar{L}$ (long-run mean) and σ (volatility)

$\bar{L}$ *long-run mean*

Economic interpretation:

The state liquidity gravitates toward in the long run, absent persistent shocks.

Calibration:

$\bar{L}$ = 1.0 corresponds to a long-run illiquidity premium near the historical average of ~3.5%.

Empirical proxy: average of secondary discount across normalization regime (~8%).

σ *volatility*

Economic interpretation:

The magnitude of period-to-period random shocks to the liquidity state.

Calibration:

σ = 0.32 produces a stationary standard deviation of ~0.337 — matching observed regime-to-regime variation.

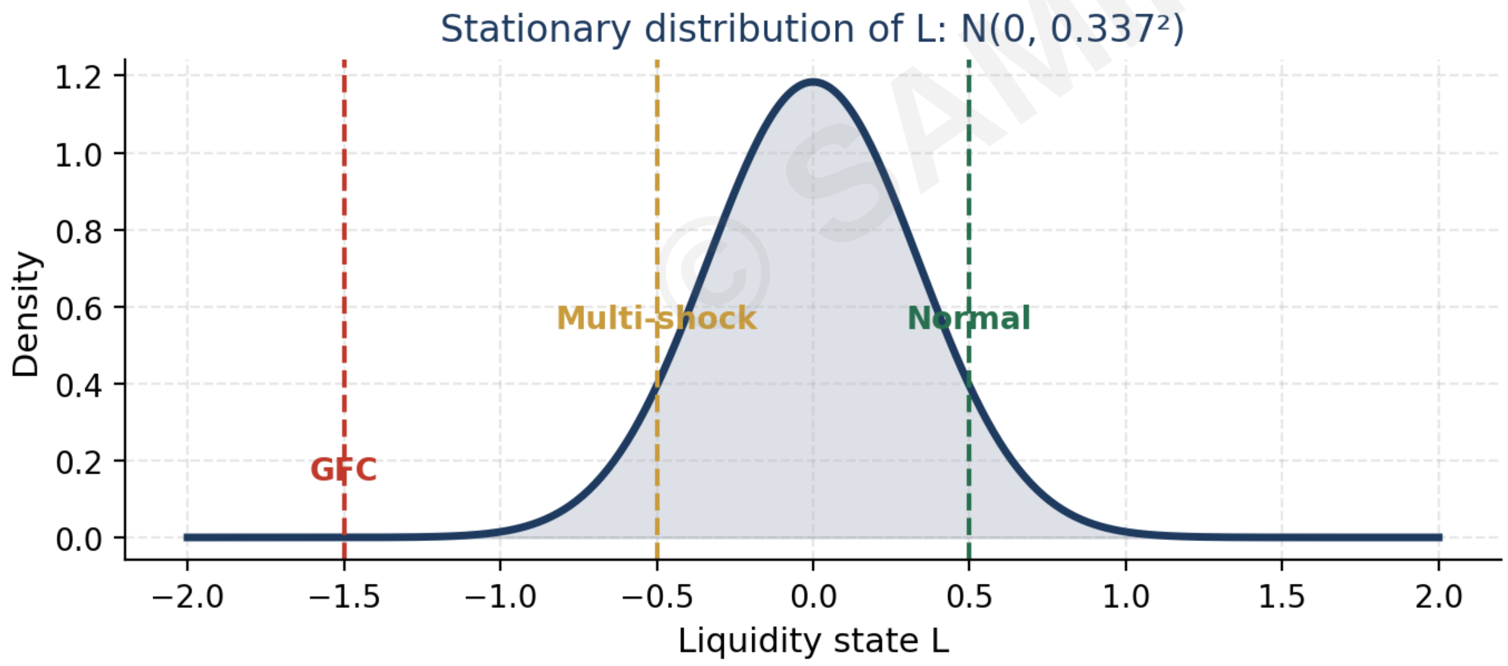
Empirical proxy: standard deviation of secondary discount across full sample (~12%).

Calibrated GE-LAV parameters: κ = 0.45/yr · $\bar{L}$ = 1.0 · σ = 0.32 · stationary std ≈ 0.337

Stationary distribution: where L(t) lives in the long run

If we run the OU process long enough, the distribution of L(t) converges to a fixed Gaussian.

$$L_\infty \sim N\left(\bar{L},\ \frac{\sigma^2}{2\kappa}\right)$$



Interpretation

Long-run mean

$E[L_\infty] = \bar{L} = 0$

Long-run variance

$Var[L_\infty] = \sigma^2/(2\kappa) \approx 0.114$

Long-run std dev

≈ 0.337

GFC depth (L=-1.5)

~3.7σ event

Frequency: GFC

<1% of time

Connection to models you may already know

The OU process has many cousins in finance. Same math, different domains:

Vasicek (1977)

Short-rate model

$dr_t = \kappa(\bar{r} - r_t)dt + \sigma dW_t$. Same SDE, modeling interest rates instead of liquidity. Closed-form bond pricing follows.

Heston (1993)

Stochastic volatility

Variance follows a CIR (cousin of OU): $dv_t = \kappa(\bar{v} - v_t)dt + \sigma\sqrt{v_t}\, dW_t$. Mean-reverting volatility.

Hull-White (1990)

Term-structure model

Extends Vasicek with time-varying parameters. Used widely in fixed income derivatives.

Hawkes processes

Self-exciting events

Used in credit defaults and limit-order book dynamics. Different SDE but similar mean-reverting intuition.

Self-diagnostic: are you ready for Track 2?

Track 2 (Researcher) does the full OU mathematics in Session 25. Including: existence and uniqueness of the SDE solution · stationary distribution proof · Karatzas-Shreve Section 5.6 treatment.

If today felt easy

Consider Track 2 (Researcher)

- ✓ You comfortable with limit operations from probability theory
- ✓ Stochastic calculus and Itô's lemma feel approachable
- ✓ You want to do GE-LAV proofs, not just apply them
- ✓ Career direction: PhD or quant research

If today felt hard (or right at the edge)

Track 1 (Practitioner) is for you

- ✓ You can apply OU intuition; deep proofs not required
- ✓ Career direction: LP, GP, regulator, consulting
- ✓ You want case-driven sessions in S25-31
- ✓ Math intuition lectures (S19-24) cover what you need

Decision by end of Session 5. Default = Track 1.

Session 4 summary

What we accomplished today

1. The OU process models $L(t)$ as mean-reverting around $\bar{L}$ with random shocks of magnitude σ
2. Three parameters: κ (speed) · $\bar{L}$ (long-run mean) · σ (vol). Calibrated: $\kappa=0.45$/yr, $\bar{L}=1.0$, $\sigma=0.32$
3. Half-life = $\ln(2)/\kappa \approx 1.5$ years (matches data)
4. Stationary distribution: Normal($\bar{L}$, $\sigma^2/2\kappa$) → GFC at $L=-1.5$ is a ~3.7σ event
5. OU has many cousins: Vasicek (rates), Heston (vol), Hull-White (term structure)

Next session

Session 5: Term structure of private capital — how the OU process pricing applies across asset lifetimes